

30.4.26 - Goldhaber experiment

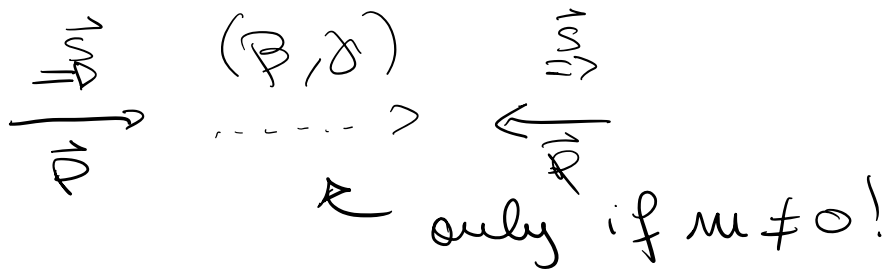
'30 Pauli $\rightarrow \exists \nu$ $n \rightarrow p + e + \nu$

1956 RC $\rightarrow$ obs. ν

1957 Wu $\rightarrow$ P violation

$\hookrightarrow$ on asymmetry $\exists$

$$h = \frac{\vec{S} \cdot \vec{p}}{|\vec{p}|} \quad \begin{matrix} \vec{p} \rightarrow -\vec{p} \\ \vec{S} \rightarrow \vec{S} \end{matrix} \Rightarrow h = -h$$

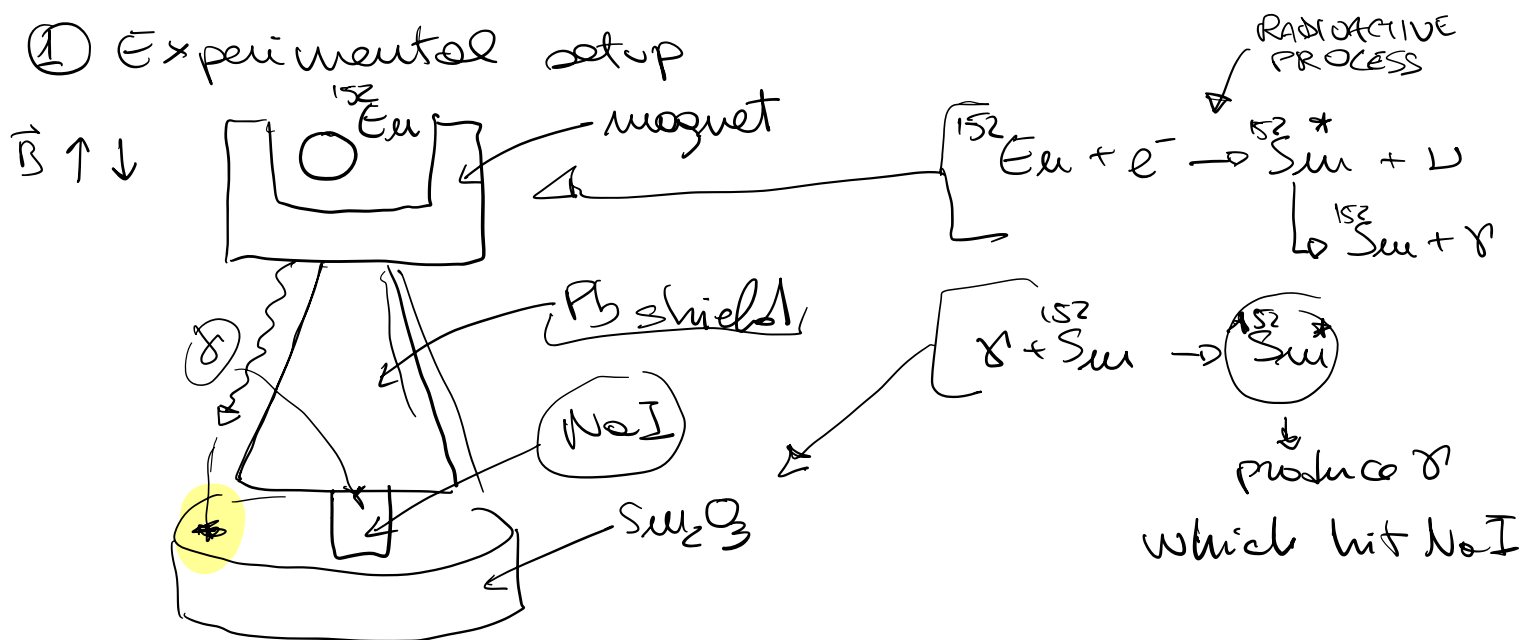


if $m=0$ ($m, \sim 0$) $\Rightarrow$ helicity is conserved

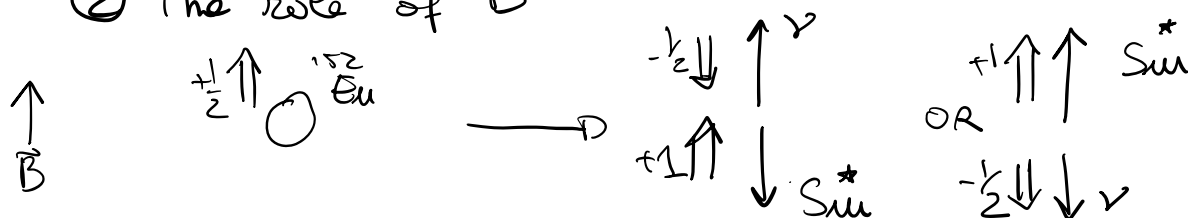
$$h(\nu) < \begin{matrix} +1 \\ -1 \end{matrix}$$

1957 $\rightarrow$ Measure $h(\nu)$! $\Rightarrow h(\nu) = -1$

① Experimental setup



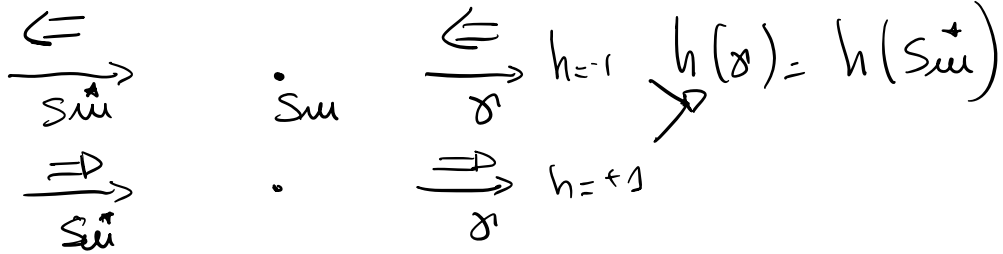
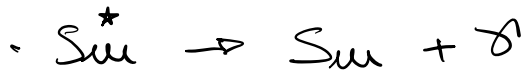
② The role of $\vec{B}$



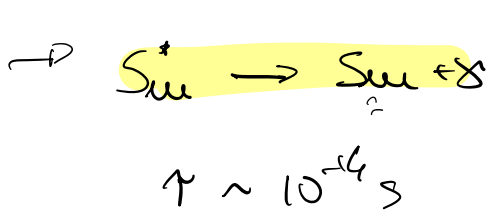
$$h(\nu) = -1 \quad \text{OR} \quad h(\nu) = +1$$

$$h(Sm^*) = -1 \quad \text{OR} \quad h(Sm^*) = +1$$

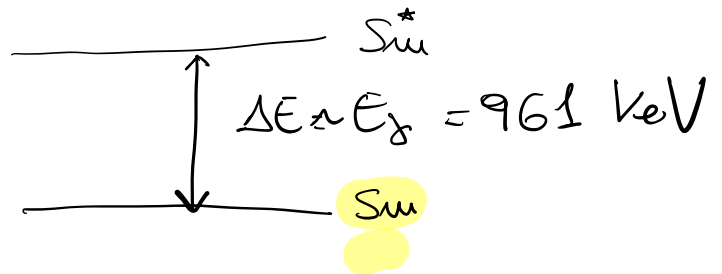
$$\Rightarrow h(\nu) = h(Sm^*)$$



③ What is the role of Sm_2O_3 ?



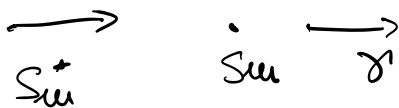
$$\Gamma = \frac{1}{\tau} \sim 10^{-2} eV$$



$$E_\gamma < 961 \text{ keV}$$

$$E_\gamma \text{ of } Sm \sim 3.2 \text{ eV} \quad \leftarrow \text{to be computed}$$

④ In the Eu decay the Sm^* gets a boost



$$E'_\gamma = E_\gamma (1 + \beta_{Sm})$$

$\beta_{Sm} \sim 10^{-6}$

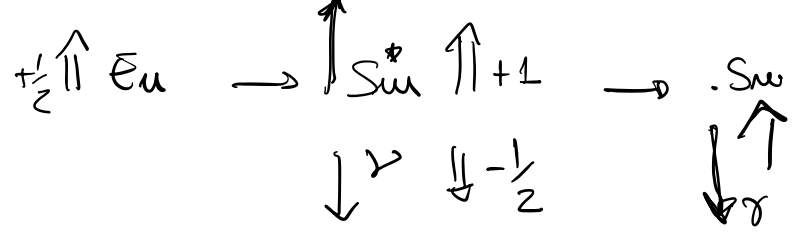
$E'_\gamma > E_\gamma$ is enough to excite Sm_2O_3

$\gamma \rightarrow$ in direction of the Sm^*

① $h(\gamma) = h(Sm^*)$

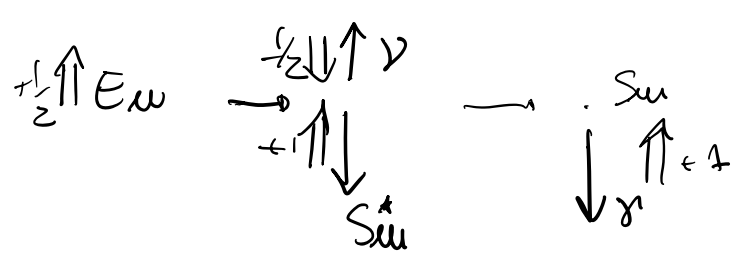
② b/c of the boost which increase E_γ to E'_γ

$\uparrow \vec{B}$
 $h = +1$



$h(\gamma) \neq h(S_{uu}^*)$
 - no boost
 NO SIGNAL

$\uparrow \vec{B}$
 $h = -1$



$h(\gamma) = h(S_{uu}^*)$
 - BOOST
 YES SIGNAL

$\downarrow \vec{B}$
 $h = +1$

$\downarrow \vec{B}$
 $h = -1$

↓

	$\vec{B} \uparrow$	$\vec{B} \downarrow$
$h = +1$	X	✓
$h = -1$	✓	X

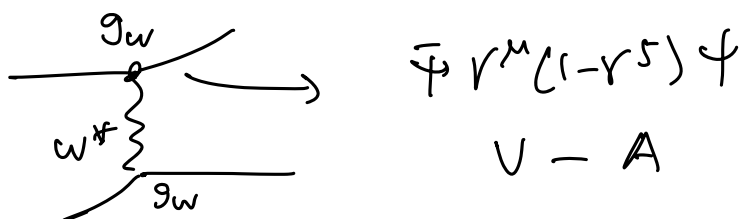
$$\Rightarrow h(\gamma) = -1$$

⑤ Do γ with $h = +1$ exist?

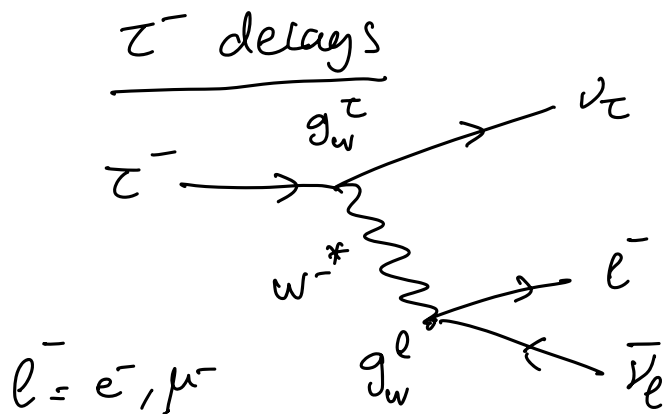
$\left\{ \begin{array}{l} h(\gamma) = -1 \text{ NO EWK} \\ q(\gamma) = 0 \text{ NO ELM} \\ \text{color}(\gamma) = 0 \text{ NO STRONG} \\ m(\gamma) \neq 0 \text{ NO GRAV} \end{array} \right.$	$\left(\begin{array}{l} h(\gamma) = +1 \\ \text{do not exist} \end{array} \right.$
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Lepton Flavor universality in weak interactions

V-A theory



Evidence that $g_w^e = g_w^\mu = g_w^\tau$?



$$m_\tau = 1.78 \text{ GeV}$$

$$m_\mu = 106 \text{ MeV}$$

$$m_e = 0.5 \text{ MeV}$$

$$\mu \sim \frac{g_w^\tau g_w^e}{m_W^2} \quad q^2 \ll m_W^2$$

$$|q| \approx m_\tau \ll m_W$$

$$\Gamma(\tau \rightarrow e \nu \bar{\nu}) \propto |M|^2 \times \rho(\tau \rightarrow e) \sim \frac{g_w^{\tau^2}}{m_W^2} \frac{g_w^{e^2}}{m_W^2} \times \rho(\tau \rightarrow e)$$

$$\sim G_F^\tau G_F^e$$

$$\tau^- \rightarrow \mu^- \bar{\nu}_\mu \nu_\tau \quad \Gamma_\mu \propto \left(\frac{g_w^\tau}{m_W}\right)^2 \left(\frac{g_w^\mu}{m_W}\right)^2 \times \rho(\tau \rightarrow \mu)$$

$$\tau^- \rightarrow e^- \bar{\nu}_e \nu_\tau \quad \Gamma_e \propto \left(\frac{g_w^\tau}{m_W}\right)^2 \left(\frac{g_w^e}{m_W}\right)^2 \times \rho(\tau \rightarrow e)$$

$$\frac{\Gamma(\tau \rightarrow \mu \nu \bar{\nu})}{\Gamma(\tau \rightarrow e \nu \bar{\nu})} = \left(\frac{g_w^\mu}{g_w^e}\right)^2 \times \frac{\rho(\tau \rightarrow \mu)}{\rho(\tau \rightarrow e)}$$

Computed analytically

$$Q_{\tau \rightarrow \mu} = m_\tau - m_\mu = 1.8 - 0.1 \text{ GeV} = 1.7 \text{ GeV}$$

$$Q_{\tau \rightarrow e} = m_\tau - m_e = 1.8 - 0.0005 \approx 1.8 \text{ GeV}$$

$$BF(\tau \rightarrow \mu) = \frac{\Gamma(\tau \rightarrow \mu)}{\Gamma_{\text{tot}}}$$

Experimentally

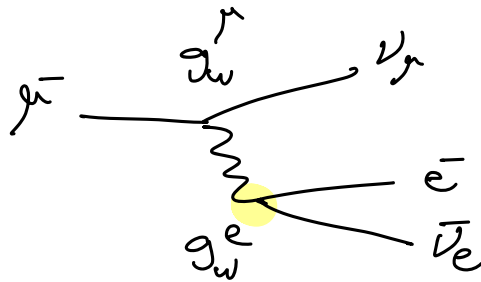
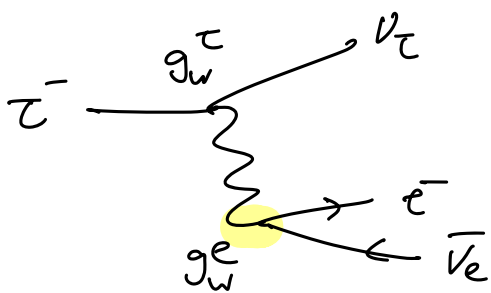
$$BF(\tau \rightarrow \mu) = \frac{\# \text{ events } \mu / \tau}{\# \text{ total } \tau \text{ produced.}}$$

$$\frac{\Gamma(\tau \rightarrow \mu)}{\Gamma(\tau \rightarrow e)} = \frac{BF(\tau \rightarrow \mu)}{BF(\tau \rightarrow e)} \frac{\cancel{\Gamma_{tot}^\tau}}{\cancel{\Gamma_{tot}^\mu}} = \left(\frac{g_w^\mu}{g_w^e}\right)^2 \times \underbrace{\frac{\rho(\tau \rightarrow \mu)}{\rho(\tau \rightarrow e)}}_{\text{computed.}}$$

measured exp.

$$\frac{g_w^\mu}{g_w^e} = 1.001 \pm 0.002 \quad \text{Same coupling} \quad g_w^e = g_w^\mu$$

$$g_w^\tau = ?$$



$$\Gamma(\tau \rightarrow e) \propto \left(\frac{g_w^\tau}{m_w}\right)^2 \left(\frac{g_w^e}{m_w}\right)^2 \rho(\tau \rightarrow e)$$

$$\Gamma(\mu \rightarrow e) \propto \left(\frac{g_w^\mu}{m_w}\right)^2 \left(\frac{g_w^e}{m_w}\right)^2 \rho(\mu \rightarrow e)$$

$$\frac{\Gamma(\tau^- \rightarrow e^- \bar{\nu}_e \nu_\tau)}{\Gamma(\mu^- \rightarrow e^- \bar{\nu}_e \nu_\mu)} = \left(\frac{g_w^\tau}{g_w^\mu}\right)^2 \times \frac{\rho(\tau \rightarrow e)}{\rho(\mu \rightarrow e)}$$

$$\frac{\Gamma(\tau^- \rightarrow e^-)}{\Gamma(\mu^- \rightarrow e^-)} = \frac{BF(\tau \rightarrow e)}{BF(\mu \rightarrow e)} \frac{\Gamma_{tot}^\tau}{\Gamma_{tot}^\mu} = \frac{BF(\tau \rightarrow e)}{BF(\mu \rightarrow e)} \frac{\tau_\mu}{\tau_\tau}$$

$$\Gamma = \frac{1}{\tau}$$

τ_τ : lifetime of τ

τ_μ : lifetime of $\mu = 2.2 \mu s$

In lab frame $N(t) = N_0 e^{-t/\tau_{LAB}} = N_0 e^{-\frac{\chi_{LAB}}{\beta \gamma \tau}}$

$$\chi_{LAB} = \beta \cdot c \cdot t_{LAB}$$

$$\tau_{LAB} = \gamma \tau$$

$$\frac{BF(\tau \rightarrow e)}{BF(\mu \rightarrow e)} \frac{\tau_\mu}{\tau_\tau} = \left(\frac{g_w^\tau}{g_w^\mu} \right)^2 \times \frac{p(\tau \rightarrow e)}{p(\mu \rightarrow e)}$$

must be measured.

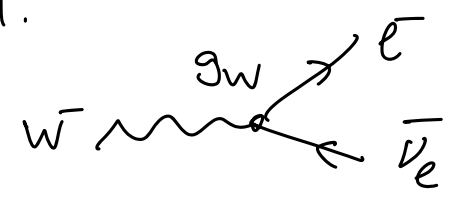
analytically computed.

From counting.

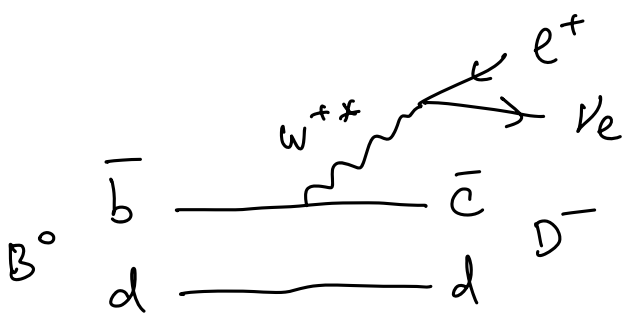
$$\Rightarrow \frac{g_w^\tau}{g_w^\mu} = 1.001 \pm 0.003$$

LHC: produce W on shell.

$$W^- \rightarrow \begin{matrix} \tau^- \nu_\tau \\ \mu^- \nu_\mu \\ e^- \nu_e \end{matrix}$$



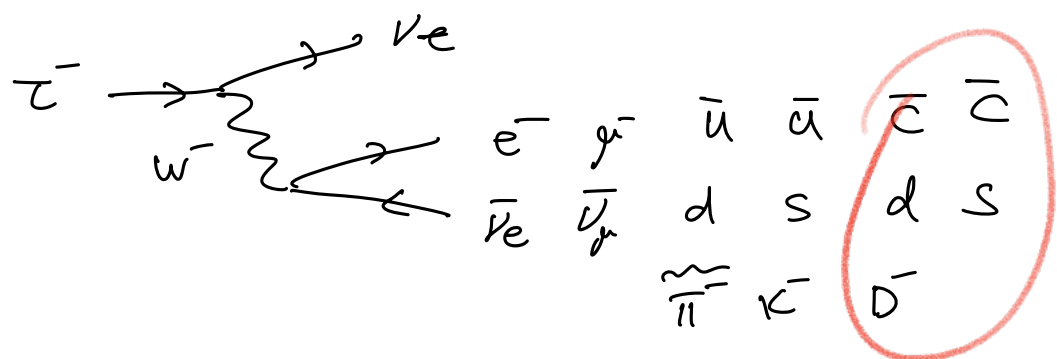
$$\begin{aligned} m_B &= 5.279 \text{ GeV} \\ m_\tau &= 1.8 \text{ GeV} \\ m_D &= 1.86 \text{ GeV} \end{aligned}$$



$$B \rightarrow \begin{matrix} \tau \nu & D^- \\ \mu \nu & D^- \\ e \nu & D^- \end{matrix}$$

$g_w^\tau = g_w^\mu = g_w^e$ universal weak coupling to W

Life time of τ estimate.



$Q < 0$ not allowed.

τ^- DECAY MODES

τ^+ modes are charge conjugates of the modes below. " $h^\pm$ " stands for $\pi^\pm$ or $K^\pm$. " ℓ " stands for e or μ . "Neutrals" stands for γ 's and/or π^0 's.

$m_\pi \approx 140 \text{ MeV}$

$m_K \approx 500 \text{ MeV}$

	Mode	Fraction (Γ_i/Γ)	Scale factor/ Confidence level
Modes with one charged particle			
Γ_1	particle $^- \geq 0$ neutrals $\geq 0 K^0 \nu_\tau$ ("1-prong")	(85.24 $\pm$ 0.06) %	
Γ_2	particle $^- \geq 0$ neutrals $\geq 0 K_L^0 \nu_\tau$	(84.58 $\pm$ 0.06) %	
Γ_3	$\mu^- \bar{\nu}_\mu \nu_\tau$	[a] (17.37 $\pm$ 0.04) %	
Γ_4	$\mu^- \bar{\nu}_\mu \nu_\tau \gamma$	[b] (3.67 $\pm$ 0.08) $\times 10^{-3}$	
Γ_5	$e^- \bar{\nu}_e \nu_\tau$	[a] (17.85 $\pm$ 0.04) %	
Γ_6	$e^- \bar{\nu}_e \nu_\tau \gamma$	[b] (1.83 $\pm$ 0.05) %	
Γ_7	$h^- \geq 0 K_L^0 \nu_\tau$	(12.03 $\pm$ 0.05) %	
Γ_8	$h^- \nu_\tau$	(11.51 $\pm$ 0.05) %	
Γ_9	$\pi^- \nu_\tau$	[a] (10.82 $\pm$ 0.05) %	
Γ_{10}	$K^- \nu_\tau$	[a] (6.97 $\pm$ 0.10) $\times 10^{-3}$	

weak interactions
are not univ.

for all flavors of quarks.

$$\Gamma_{\text{tot}}^\tau \approx \Gamma_\mu + \Gamma_e + \Gamma_{u\bar{d}} \times 3$$

